1. Application Context

In many African countries, including Lesotho, access to affordable healthcare is limited by infrastructure, cost, and distance from medical facilities. Early diagnosis of diseases such as tuberculosis (TB) is often delayed, as patients may not realize they are sick until symptoms are advanced. The Fourth Industrial Revolution (4IR) provides opportunities to address these challenges through artificial intelligence (AI) and mobile technologies. One promising area is the use of AI for cough-based TB detection. Research shows that cough sounds contain distinctive acoustic patterns which can be analyzed to provide preliminary screening (Botha et al., 2018; Larson et al., 2011). Since most people already own smartphones with microphones, this approach enables low-cost, accessible diagnostics using devices that people already carry.

This project therefore focuses on designing a custom RISC-V processor optimized for cough recognition workloads. By targeting signal analysis, memory access, and pattern comparison, the processor will support real-time TB screening on low-cost AI-enabled mobile phones. Such a solution has the potential to improve healthcare access and reduce the burden of disease across Africa.

2. Representative Applications

Application	Short Description	Key Computational Tasks	Notes (Relevance)
Cough-Based TB Detection	Analyze cough sounds for TB screening	FFT/MFCC feature extraction, array operations, pattern matching	Relevant for rural healthcare, low-cost screening
Feature Extraction	Transform raw cough audio into frequency-domain features	FFT, MFCC computation, array storage	Fundamental workload; drives processor arithmetic and memory demands
Reference Comparison	Compare extracted cough features with stored TB-positive patterns	Array comparisons, conditional branching, loops	tests control flow, memory access patterns, and decision logic
Diagnostic Output	Provide feedback based on similarity results	Conditional checks, output signaling	Emphasizes low-latency branching and interface with display/output modules

3. Workload Characteristics

The program should convert the raw cough sound into frequency-domain data, using transformations such as the Fast Fourier Transform (FFT) or Mel-frequency cepstral coefficients (MFCCs). The extracted features will be stored in arrays in memory.

The system should be able to load stored cough feature data (representing reference TB-positive cases) from memory into registers for comparison.

The program should compare the features of the current cough with the stored reference features. This may involve conditional checks (if/else) and iterations (for loops) to measure similarity.

Based on the comparison, the program should output a result, for example, "Possible TB detected, seek medical testing" or "Cough not consistent with TB." Furthermore, in C++ this can be implemented using *cout* statements, but in hardware, it will translate into specific output signals.

4. Initial Insights for Processor Design

- Efficient support for Fast Fourier Transform and feature extract.
- Optimized memory hierarchy for frequent array loading and comparison.
- Low-latency branch handling for classification and decision-making.
- Energy-efficient design suitable for low-cost mobile phones in resource-constrained environments.
- ISA extensions for signal processing and AI workloads (e.g., pattern matching).

References

World Health Organization (WHO).
Global Tuberculosis Report 2023. Geneva: WHO, 2023.

Botha, G., Theron, G., Warren, R., Klopper, M., et al.
"Detection of tuberculosis by cough sound analysis: A proof of concept." *The International Journal of Tuberculosis and Lung Disease,* vol. 22, no. 8, pp. 910–918, 2018.

Larson, E. C., Lee, T., Liu, S., Rosenfeld, M., & Patel, S. N.
"Accurate and Privacy Preserving Cough Sensing Using a Low-Cost Mobile Microphone." *Proceedings of the 13th International Conference on Ubiquitous Computing (UbiComp).* ACM, 2011.

RISC-V International.
The RISC-V Instruction Set Manual, Volume I: User-Level ISA, Version 20191213. RISC-V Foundation, 2019.